

Power BI Dashboard - Final Package README

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Real Estate AI Decision Intelligence Platform — Power BI Dashboard (Final Package)

This package is the final, consolidated Power BI deliverable for the platform's 10-phase architecture (Master A-Z Development Prompt). It merges two things into one:

1. **The generated data layer and reference code** (`phase10a_source/`) — the 12 Power BI data marts, the Python code that built them, the source CSVs each mart was derived from, automated tests, and full documentation.
2. **The actual, hand-built Power BI report** (`Real_estate_dashboard.pbix`) — built directly in Power BI Desktop from those 12 marts, across 7 pages, plus screenshots of the finished result.

What's in this package

<code>Real_estate_dashboard.pbix</code>	<- Open this in Power BI Desktop
<code>dashboard_screenshots/</code>	<- 7 PNG screenshots of the finished report
<code>phase10a_source/</code>	<- Full backend: code, marts, tests, docs
<code>powerbi_build_kit/</code>	<- Build guide, DAX measures, color theme

`Real_estate_dashboard.pbix`

The actual Power BI file. Open it directly in Power BI Desktop — no setup required, all data is embedded. Contains 7 report pages (see screenshots below for what each one looks like).

`dashboard_screenshots/`

File	Page
<code>01_executive_overview.png</code>	Portfolio KPIs, decision distribution donut, invest rate
<code>02_market_locality_intelligence.png</code>	Top-15 price growth chart, demand-vs-supply scatter, city filter
<code>03_property_investment.png</code>	Property table, valuation-gap chart, decision filter
<code>04_financial_risk.png</code>	IRR/NPV/DSCR table, Monte Carlo scenario chart, VaR/CVaR cards, UNCALIBRATED disclaimer
<code>05_geo_intelligence.png</code>	Locality map, accessibility/transit/commercial scores, flood-data-gap card
<code>06_decision_intelligence.png</code>	Component-score breakdown, gate pass-rate, per-property decision table, reasons
<code>07_data_quality_monitoring.png</code>	Full governance status table, rejected-model count, status distribution

`phase10a_source/`

The complete backend that produced the 12 CSV data marts the `.pbix` file was built from —

`src/marts/build_marts.py`, the source CSVs under `sources/phase6` through `sources/phase9`, the `data_marts/*.csv` outputs themselves, the automated test suite (`src/marts/test_phase10a.py`), and full phase documentation (`docs/phase10a_report.md`, `docs/limitations.md`).

`powerbi_build_kit/`

Reference material used to build the report: a step-by-step build guide (`POWER_BI_BUILD_GUIDE.md / .pdf`), ready-to-paste DAX measures (`DAX_measures.md`), and the custom Power BI color theme (`power_bi_theme.json`) matching the platform's dark gold/green/red palette.

Independent recheck performed before this package was assembled

Since this package combines two separately-delivered pieces (the generated marts and the hand-built `.pbix`), the following was re-verified end to end:

1. **ROI calculation fix confirmed present.** `phase10a_source/src/finance/engine.py` was checked and confirmed to contain the corrected `roi_total_holding_period` formula (net return, not gross cash multiple) — the fix found during a prior audit of Phase 6 and traced through Phases 7, 9, and this phase. Verified directly: property 2731's Base-case ROI in `mart_04_financial_risk.csv` is **123.77%**, consistent with its 17.85% IRR compounded over a 5-year hold (not the pre-fix 223.77%).
2. **Automated test suite re-run: 9/9 passing.** `phase10a_source/src/marts/test_phase10a.py` was executed fresh inside this package — covering the ROI regression check, cross-scenario IRR/ROI monotonicity, the property-1 NaN-locality-name regression check, portfolio-KPI-vs-decision-source consistency, and structural checks that no financial/risk formula is recomputed in the mart-building code.
3. **Data-mart consistency confirmed byte-for-byte.** Every one of the 12 CSV files under `phase10a_source/data_marts/` was diffed against the copy that was actually used to build `Real_estate_dashboard.pbix` (originally supplied separately as the “build kit” data). All 12 files are identical — confirming the `.pbix` was built from the correct, already-fixed data, not a stale copy.
4. **All 7 dashboard screenshots reviewed.** Each of the issues found and fixed during the interactive build process is confirmed resolved in the final screenshots: Page 3's valuation-gap chart uses a categorical (not continuous) axis; Page 4's probability/VaR/CVaR cards use Average (not Sum) aggregation and carry the UNCALIBRATED disclaimer text box; Page 5's flood-risk-data-gap card shows a real percentage (100%) with an explanatory note about partial transit/commercial score coverage; Page 6's decision table shows one row per property with genuinely distinct scores (not grouped/summed by decision category); Page 7's column headers are correctly labeled and the Rejected Model Count card correctly reads 2 (not the row-count of 1).

No further issues were found. This package is internally consistent and ready to present.

Key facts for a resume, portfolio page, or interview

- **26 properties fully evaluated** end-to-end through a 9-phase decision pipeline (data → SQL analytics → EDA → 7 ML models → evidence/RAG → financial engine → Monte Carlo risk → geo scoring → multi-agent decision policy), surfaced here as a 7-page interactive Power BI report.
- **2 of 7 ML models were rejected** after independent validation showed near-random predictive power ($R^2 \approx 0$ for days-on-market, ROC-AUC ≈ 0.52 for sale probability) — shown transparently on the Data Quality page rather than hidden.
- **Every Monte Carlo/VaR/CVaR figure is explicitly labeled UNCALIBRATED** — a documented judgement call, not fitted to real historical outcomes. This is stated directly on the Financial & Risk page, not buried in a footnote.
- **Data gaps are shown honestly, not imputed.** The Geo Intelligence page states plainly that flood-risk data is unavailable for 100% of localities, and that transit/commercial accessibility scores only exist where the underlying infrastructure records do.
- **Every decision is explainable.** Page 6 lets you click any property and see exactly which of 5

weighted components (market, prediction, financial, risk, geo) drove its score, which governance gates it passed, and the specific stated reasons behind the INVEST/HOLD/AVOID/INSUFFICIENT call.

- **One real, systematic bug (the ROI calculation) was found during independent audit and fixed across every phase it had propagated into** — a concrete example of QA discipline applied to a multi-stage data pipeline, not just a one-off model built and shipped.
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How to open

1. Extract this package.
2. Double-click `Real_estate_dashboard.pbix` to open it directly in Power BI Desktop — all data is embedded, no data source reconnection needed.
3. To see the source data pipeline or re-run the tests, see `phase10a_source/README_PHASE10A.md`.